

Question 1: Chi-Square Test

a)

H_0 : The observed frequencies follow the expected distribution, which is the coin is fair.

H_a : The observed frequencies did not follow the expected distribution.

b)

$$N = 240, \quad E_2 = 240 - (30 + 90 + 30) = 90$$

Outcome	Expected	$\frac{(O-E)^2}{E}$
0 head: 24	30	$\frac{(24-30)^2}{30} = 1.20$
1 head: 108	90	$\frac{(108-90)^2}{90} = 3.60$
2 heads: 95	90	$\frac{(95-90)^2}{90} = 0.28$
3 heads: 23	30	$\frac{(23-30)^2}{30} = 1.63$
Total: 240	240	$\chi^2 = 6.71$

c)

- Degrees of freedom: $df = k - 1 = 4 - 1 = 3$
- Critical value: $df = 3, \alpha = 0.05, \chi^2_{crit} = 7.81$
- Decision and Interpretation:** Fail to reject H_0 , since $6.71 < 7.81$.

Therefore, there is no significant difference between the O and E frequencies. As expected, the coins behave.

Question 2

a)

$$E = \frac{(\text{row total} \times \text{column total})}{N}$$

Vaccinated:

- got disease: $E = \frac{(51 \times 26)}{96} = 13.81$
- no disease: $E = \frac{(51 \times 70)}{96} = 37.19$

Not vaccinated:

- got disease: $E = \frac{(45 \times 26)}{96} = 12.19$
- no disease: $E = \frac{(45 \times 70)}{96} = 32.81$

Question 3

b)

$$\chi^2 = \frac{(9-13.77)^2}{13} + \frac{(42-37.23)^2}{37.23} + \frac{(17-12.33)^2}{12.33} + \frac{(28-32.77)^2}{32.77}$$
$$\chi^2 = 1.65 + 0.61 + 1.86 + 0.96 = 4.81$$

c)

The vaccination and disease relationship between them are significantly occurrence.